

AJCR Mathematical Diagnosis for Human Review

Human mathematician handoff

24 August 2026

Diagnosis. The route is not stalled by formal transport alone. Two producer theorems are missing: (i) a functor-level representability theorem for the exact finite-stage carrier `P.gluedOver`; and (ii) orbit-affine/projective geometry for that same carrier over an arbitrary finite Galois stage. The later descent and Jacobian steps are conditional downstream work.

This is a mathematics-only comparison with the supplied execution plan and supervision note. File pointers below identify the definitions and theorem statements where each claim lives.

`MainProjects/Algebraic-Jacobian-Challenge/informal/Lean_Algebraic_Jacobian_Complete_Execution_Plan.pdf`

`MainProjects/Algebraic-Jacobian-Challenge/informal/AJCR_Runs_121_122_123_Supervision_Note_2026-08-07.pdf`

1. Route being formalized

rank-one Picard/divisor chart

- > family-level Abel isomorphism
- > translated charts over a separably closed field
- > `Pic0` represented over that field
- > finite-stage glued carrier `P.gluedOver`
- > finite-Galois quotient and descent
- > `Pic0` over the original field
- > Jacobian

The candidate carrier is defined at:

`MainProjects/Algebraic-Jacobian-Challenge-Rebuild/AlgebraicJacobian/Picard/Pic0FiniteStageGluedOver.lean:1`

The decomposition of the finite-stage descent is recorded at:

`.archon-horizon/roadmap/items/AJCR.review-plan.p7-galois-descent.yaml`

2. Established mathematics

Stage	Statement and pointer
Rank-one chart	Established The good line-bundle locus and matching divisor locus are defined over arbitrary test schemes. <code>Pic0CriticalPath.lean:205</code>
Family Abel map	Established The evaluation divisor gives a natural two-sided inverse, not only a fieldwise bijection. <code>Pic0CriticalPath.lean:618</code>
Separably-closed cover	Established Translates of the rank-one chart cover Picard classes over a separably closed field. <code>Pic0CriticalPath.lean:640</code>
Separably-closed <code>Pic⁰</code>	Established A scheme and a representing universal class exist over the separably closed field. <code>Pic0SepClosedRepresentable.lean:443</code>
Local universal class	Established The pinned class has both chart-to-overlap restriction equalities and is packaged as <code>Pic0FiniteStageUniversalAtlasClass</code> . <code>Pic0FiniteStageUniversalClass.lean:119</code>
Scheme comparison	Established <code>gluingGluedIso</code> and <code>finiteStageBaseChangeIso</code> compare the scalar-extended finite-stage glue with the separably-closed atlas glue as schemes. <code>Pic0FiniteStageGluedComparison.lean:239</code>

The original rank-one family producer is therefore no longer the obstruction, and local overlap compatibility of the universal class is already present.

3. The two mathematical producer gaps

GAP A — finite-stage Yoneda representability

Missing statement. For every test scheme T over $\text{Spec } k$, construct a precomposition-natural equivalence between maps to $P.\text{gluedOver}$ and $\text{picOfTypeFuncor } C \ T$. Equivalently, produce the following Lean proposition with the exact carrier and no additional binder:

```
(picOfTypeFuncor ((baseChange K P.N.1).obj C)).RepresentableBy P.gluedOver
```

Why the current objects do not prove it. The glue package gives charts, overlaps, and a glued scheme. `finiteStageBaseChangeIso` and `gluingGluedIso` give an isomorphism of schemes after base change. Neither statement identifies maps from an arbitrary T with Pic0 families on T ; that identification is the missing natural `homEquiv`. The roadmap names this target explicitly and records that no producer for it exists.

```
.archon-horizon/roadmap/items/AJCR.review-plan.p7-galois-descent.universal.yoneda.yaml
```

```
.archon-horizon/roadmap/items/AJCR.review-plan.p7-galois-descent.representability.finite-stage.yaml
```

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/blueprint/src/chapters/DivisorScheme.tex:1846
```

GAP B — geometry of the exact finite carrier

Missing statement. Prove `FiniteInAffine P.glueData.glued`, or an equivalent finite-dimensional projective-space immersion/projectivity theorem, over the arbitrary finite Galois stage used by P . The statement must concern the same carrier that is supposed to represent the functor.

What is conditional already. `PicOfFiniteStageOrbitAffine.lean` has producers after consuming `rep`; some require an algebraically closed field plus connectedness or irreducibility, while another requires an explicit immersion or projectivity input. The stable-cover wrappers consume the same hypotheses. There is no unconditional arbitrary-field producer before Gap A. This is a genuine geometric gap unless the reviewed finite-type/proper/group facts imply the missing hypotheses.

```
.archon-horizon/roadmap/items/AJCR.review-plan.p7-galois-descent.orbit-affine.exact-carrier.yaml
```

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/AlgebraicJacobian/Picard/PicOfFiniteStageOrbitAffine.lean:56
```

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/AlgebraicJacobian/Picard/PicOfFiniteStageStableAffineCover.lean:43
```

Dependency order. Gap B's orbit-affine producers consume Gap A's `RepresentableBy` certificate. They cannot create that certificate. This is why strengthening orbit lemmas before proving the finite-stage universal property does not move the route forward.

4. What is not a new mathematical obstruction

LIKELY FORMAL TRANSPORT — global class gluing

The chart and overlap classes already satisfy both restriction equations. The remaining statement is to pass them through the multicoequalizer and show that pullback along every test map gives the class selected by Gap A. This should preserve the one pinned universal class; introducing a second class would change the route.

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/AlgebraicJacobian/Picard/PicOfFiniteStageUniversalClass.lean:135
```

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/AlgebraicJacobian/Picard/PicOfFiniteStageGluedComparison.lean:284
```

LIKELY FORMAL TRANSPORT — finite-Galois descent

The blueprint already states the representation-preserving quotient/descent theorem. Once Gap A and Gap B supply its hypotheses, the remaining work is to carry the same scheme, semilinear action, universal class, cocycle, and naturality through that theorem.

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/blueprint/src/chapters/DivisorScheme.tex:1777
```

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/blueprint/src/chapters/DivisorScheme.tex:1846
```

LIKELY FORMAL TRANSPORT — Jacobian handoff

The descended `RepresentableBy` datum should pass unchanged through `PicRepDatum`, `JacobianData`, and the final Jacobian statement. A second representing scheme or universal class would be a change of mathematics, not packaging.

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/AlgebraicJacobian/Picard/PicOfFiniteGaloisJacobianData.lean:1
```

```
MainProjects/Algebraic-Jacobian-Challenge-Rebuild/AlgebraicJacobian/Challenge.lean:1
```

5. Why the route keeps reaching the same boundary

The current strategy accumulates restriction, overlap, diagram, and base-change identities for a finite atlas. Those facts show that $P.\text{gluedOver}$ is a plausible model and that its scalar extension matches the separably-closed model. They do not prove the universal assertion

$$\text{for every } T, \quad \text{Maps}(T, P.\text{gluedOver}) \text{ is naturally } \text{Pic}^0(T).$$

That assertion is the first producer theorem consumed by every later descent construction. The orbit-affine lemmas then consume the missing rep ; they cannot create it. Thus local identities can all be correct while the route still stops at Gap A.

6. Mathematical conclusion

The unresolved route is concentrated in two concrete claims: (A) finite-stage functor-level representability by $P.\text{gluedOver}$; and (B) exact-carrier orbit geometry over an arbitrary finite Galois stage. These are the places where a new mathematical idea, a stronger hypothesis, or a different carrier may be required.

Class gluing, effective finite-Galois descent, and the Jacobian handoff are conditional downstream work. They should be attempted only after the two producer statements are mathematically justified, and they should preserve the same representing scheme and universal class.

The useful review decision is therefore concrete: validate or reject Gap A and Gap B as stated. If either theorem cannot be formulated for the exact carrier without extra hypotheses, the present route needs redesign; further local overlap identities will not resolve that issue.